

# Hima — QA Release Testing SOP & Roadmap

Standard operating procedure for testers: what to test **before** release and **after** release  
Innovfix Pvt. Ltd. · Hima App · v1 · 2026-07-14 · Owner: Yuvanesh (Tech Lead)

**How to use this doc.** For every release: run **Phase 1–4 (before release)** and only ship when the sign-off gate passes; then run **Phase 5 (after release)** on production. **BEFORE** = pre-release on staging/APK. **AFTER** = post-release on live prod. **MUST-PASS** = a failure blocks release.

## 1. Testing Phases (the process)

| Phase                            | When                | What                                                                             |
|----------------------------------|---------------------|----------------------------------------------------------------------------------|
| 0 · Build sanity                 | New APK arrives     | Installs, opens, no crash on launch, correct version number, login screen loads. |
| 1 · Regression                   | Before release      | Every module's core flows still work (Section 3 checklists).                     |
| 2 · New-feature                  | Before release      | Deep-test whatever changed in THIS release (from the branch/changelog).          |
| 3 · Critical-path                | Before release      | Money + calls deep test (Section 4) — zero tolerance.                            |
| 4 · Sign-off gate                | Before release      | No Sev-1/Sev-2 open. Section 6 checklist signed.                                 |
| 5 · Post-release smoke + monitor | Right after release | Prod smoke on 2–3 real devices + watch live metrics for 2–4h (Section 5).        |
| 6 · Rollback watch               | First 24h           | If a rollback trigger hits (Section 5), revert/hotfix immediately.               |

### Environments & test accounts (set up once)

- Devices:** ≥2 Android phones (1 low-end, 1 high-end), on WiFi **and** mobile data. Test the **Play Store / Internal Testing** build, not only sideload (sideload can pass while Play build fails — always confirm on the real distribution channel).
- Accounts:** a **male caller** (with test coins), a **female creator** (verified, with bank/UPI + PAN), and a **fresh new user** (for onboarding). Keep a second creator for block/DND pairing tests.
- Money:** use small real amounts (₹1–₹49) for real gateway tests — never assume; a coin credit must be verified end-to-end.

## 2. Bug severity & the release gate

| Severity                 | Examples                                                                                                                                             | Blocks release?                       |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Sev-1</b><br>Critical | Coins credited without payment / free coins · double billing · double payout · app crash on launch · calls don't connect · can't log in · money lost | 🔴 <b>YES — hard block</b>             |
| <b>Sev-2</b><br>Major    | A gateway fails · withdrawal stuck · matcher connect-rate drops · KYC/verify broken · chat not delivering                                            | 🔴 <b>YES</b>                          |
| <b>Sev-3</b><br>Minor    | UI glitch, wrong text, non-blocking edge case                                                                                                        | 🟡 <b>Ship with a follow-up ticket</b> |

## 3. BEFORE RELEASE — module-by-module roadmap **BEFORE**

### 3.1 Login · Auth · Onboarding

| ✓ Test                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> New user: enter mobile → OTP received → verify → profile setup → lands on home.           |
| <input type="checkbox"/> Wrong OTP rejected; resend OTP works; OTP expiry handled.                                 |
| <input type="checkbox"/> Existing user login + logout + re-login (token refresh) works.                            |
| <input type="checkbox"/> Profile update: name (one-time name-change rule), avatar, interests (optional), language. |

#### ✓ Test

- Gender selection correct; male vs female home screens differ correctly.
- AI onboarding chat triggers for new male users; icebreaker shows.
- Force-update: a below-min-version user is prompted/blocked per `individual_app_update`.
- Delete account flow works; deleted user can't log back in.
- **NEW** Keyboard visibility: after entering the mobile number, the **mobile number field is fully visible** above the keyboard (not hidden); and on the OTP step the **6-digit OTP boxes are fully visible** above the keyboard. Both steps, low-end + high-end device. (QA #4)

### 3.2 Creator verification / KYC

#### ✓ Test

- Voice verification: record voice → female auto-approve / male-voice reject.
- PAN KYC (Cashfree Secure-ID, PaySprint fallback): valid PAN passes, invalid rejected.
- Bank + UPI details save + validate; can't withdraw before verification.

### 3.3 Calls (audio + video) **MUST-PASS parts**

#### ✓ Test

- **Random match:** caller taps random → matched to an online creator in his language → rings.
- **Connect:** creator answers → Agora audio/video connects both ways → timer starts ( `started_time` ).
- **Billing:** coins deduct per minute from caller; creator income accrues correctly; amounts match the rate.
- **No double-billing** (rapid reconnects / weak network don't charge twice).
- **Coins exhaust:** server force-ends the call when caller's coins hit 0.
- **Direct call** to a specific creator; recent/favourite call works.
- **Reject:** creator rejects → caller sees "busy/unavailable"; 3-reject cool-down applies.
- **Missed call:** unanswered → `missed_call` count + push to creator.
- **Video & audio both;** switch between; low-network behaviour (call doesn't hang forever).
- Call ends cleanly on either side; end reason recorded; earnings shown to creator.
- **NEW Network drop while connecting:** caller's network dies mid-ring → the **creator's ring auto-ends** within ~45s (no connected-but-silent call that has to be hung up manually). (QA #2)
- **NEW Ring stops on cancel — app closed:** creator's app is closed/backgrounded and rings → caller cancels → the ringing/vibration **stops immediately** (no ghost ring to dismiss by hand). (QA #6)
- **NEW Call history label:** a connected + completed call always shows its **duration** (even a very short call) in Call History — it is **never** labelled "Missed". Genuinely unanswered calls still show "Missed". (QA #10)
- **NEW No duplicate/ghost calls on reopen:** after the creator closes and reopens the app, previously received incoming/missed calls do **not** re-appear or re-ring as new incoming calls (queued pushes for already-ended calls are dropped). (QA #7)

### 3.4 Do-Not-Disturb & Blocking

#### ✓ Test

- Creator turns DND on → she gets NO incoming calls/chat pings → shows unavailable → stays on until she turns it off.
- Call-block: blocked caller can't call her at all; matcher never pairs them again.
- Chat-block: blocked caller can't message; existing chat stops.
- Both honoured silently (no "you are blocked" popup) across call + chat + matcher.
- **NEW** After blocking a contact, that conversation **moves to the bottom** of the chat list (and shows the Blocked badge). (QA #8)

### 3.5 Coins & Payments MUST-PASS

**Zero tolerance:** coins must be credited ONLY after a verified real payment, exactly once. Test EACH live gateway with a small real amount.

#### ✓ Test

- ☐ **Cashfree:** buy a pack → pay → coins credited exactly once; check `transactions.add_coins`.
- ☐ **PhonePe:** same end-to-end.
- ☐ **Razorpay / UPI link:** same end-to-end.
- ☐ **Payment FAIL / cancel:** NO coins credited; balance unchanged; clean error to user.
- ☐ **Webhook replay / forged:** a repeated or unverified webhook does NOT re-credit or free-credit coins.
- ☐ Offers, trial coins, coupons apply the correct discounted price + coins.
- ☐ Gateway switch/maintenance flag: when a gateway is off, the app falls back correctly.

### 3.6 Autopay / Subscription

#### ✓ Test

- ☐ ₹1 trial sheet shows to never-subscribed; mandate created (UPI, GPay-compatible).
- ☐ Daily coin claim works; recurring ₹299 charge fires; subscription status correct.
- ☐ Cancel subscription works; CHARGE-fail vs AUTH-fail push messages correct.

### 3.7 Withdrawals / Payouts MUST-PASS

**Zero tolerance:** a creator must be paid the correct amount exactly once — never double-paid, never a phantom deduction.

#### ✓ Test

- ☐ Request withdrawal → balance deducts → Cashfree payout (IMPS) → status Pending→Paid → money reaches bank.
- ☐ **No double-pay** (rapid taps / retries don't pay twice).
- ☐ Failed/rejected transfer → balance rolled back correctly.
- ☐ PAN gate + TDS applied where required; withdrawal blocked without PAN.
- ☐ Min-amount, bank-detail-missing, and reversal (auto-refund coins) edge cases.

### 3.8 Creator earnings, tiers & bonuses

#### ✓ Test

- ☐ Income from calls shows correctly on creator dashboard; balance + `total_income` match.
- ☐ Tier/score updates; call-bonus tiers pay the right bonus; icebreaker prompts show.
- ☐ Creator warnings issue + auto-disable + auto-unblock work; star-creator rates apply.
- ☐ **NEW** **Call-duration match:** the creator's **Call History duration = Earnings/Transactions duration = the real connected talk time** (measured answer→hangup, with no ring/setup time included). (QA #12)
- ☐ **NEW** **Every connected call is credited:** a call that connected ( `started_time` set) credits the creator for its real talk time **even if the caller's app didn't cleanly report the end** (force-close / network drop) — no answered call is left with ₹0 income. (QA #11)

### 3.9 Chat & Social (real-time)

#### ✓ Test

- ☐ Send/receive message in real time (both directions); single→blue ticks (read receipts).
- ☐ Reactions add/update; message delete (archived server-side) + reflects for both.



✓ **Test**

- ☐ Bad-word message filtered/handled; presence/typing indicators show.
- ☐ Attachments upload; gifts send + deduct coins + credit creator; friend requests flow.
- ☐ Open-chat-from-notification deep-links to the right thread.
- ☐ **NEW** Image and audio **attachment options are removed** from the chat composer (emoji available in their place). (QA #3)

### 3.10 Notifications

✓ **Test**

- ☐ Push received (FCM + OneSignal): incoming call, new message, friend request, creator-online.
- ☐ Tapping a notification opens the correct screen; DND suppresses pings.
- ☐ Notification click/conversion tracking records (if part of this release).
- ☐ **NEW** Tapping a **video**-call notification starts a **video** call (not audio) when both call types are enabled. (QA #9)
- ☐ **NEW** Rejected calls are **not** shown/pushed as "missed"; missed-call notification text shows the correct call status (not both audio & video). (QA #5)

### 3.11 Growth, Games & misc

✓ **Test**

- ☐ Install referrer / tracking link / deep link attribute correctly (if changed).
- ☐ IPL rooms create/join/leave/react; Ludo invite + play (if in release).
- ☐ Reports/ratings/reviews; app-store rating prompt triggers per rules.

### 3.12 Admin panel (backend) **BEFORE**

✓ **Test**

- ☐ **NEW** Block a user/creator with a reason → reopen the account → the **block reason and blocked status/icon are still shown** (retained for audit). (QA #1)

## 4. CRITICAL MUST-PASS scenarios (money + calls) **MUST-PASS**

These four must be re-verified **every release**, end-to-end, on the real Play build. A single failure = **do not ship**.

1. **Pay → get coins, once.** Real payment on each live gateway → exact coins credited once. Fail/forged → no credit.
2. **Call → correct billing.** Connect → per-minute deduct + creator income correct → no double-bill → force-end on 0 coins.
3. **Withdraw → paid once.** Payout reaches bank, correct amount, no double-pay, rollback on failure.
4. **Matcher + safety.** Random call connects (~normal rate), and DND/blocked pairs are never matched.

## 5. AFTER RELEASE — production verification & monitoring **AFTER**

### 5.1 Smoke test on prod (first 30 min, 2–3 real devices)

✓ **Test on the LIVE app**

- ☐ Fresh install from Play → open → login/OTP works.
- ☐ One real random call connects both ways.
- ☐ One real small coin purchase credits coins.
- ☐ One real chat message delivers + blue ticks.

✓ **Test on the LIVE app**

- ☐ A test withdrawal request is accepted (status pending).

## 5.2 Watch these live metrics for 2–4 hours

| Metric                             | Healthy                     | Rollback trigger                               |
|------------------------------------|-----------------------------|------------------------------------------------|
| <b>Connect rate</b> (started_time) | ~45–52%                     | Sudden drop >10 pts vs same slot yesterday     |
| <b>Revenue</b> (add_coins/hr)      | ≈ prior day same hour       | Sharp fall vs baseline (not a demand dip)      |
| <b>Gateway success %</b>           | Cashfree ~56%, PhonePe ~51% | Any gateway drops >10 pts                      |
| <b>Payout %</b>                    | ~40%                        | Spikes (income leak / ghost calls)             |
| <b>Crashlytics crash-free</b>      | >99%                        | New crash cluster after release                |
| <b>Error log</b> (laravel.log)     | baseline noise              | New error spiking (e.g. class-not-found, 500s) |

**Version-aware check:** compare the NEW version's users (by `current_version`) vs the stable version on connect rate, ARPU, and gateway success — a bad release shows as the new version underperforming (like v1109 did). Roll out in staged % and watch before 100%.

## 5.3 Rollback / hotfix criteria

- Any **Sev-1** in production (money loss, free coins, double-pay, mass crash, calls not connecting) → **halt rollout + hotfix/rollback immediately**.
- Connect rate or revenue drops sharply and it's traced to the release (not external) → roll back the staged %.
- A gateway or payout breaks → flip its kill-switch ( `gateway_config` ) and fix.

## 6. Sign-off checklist (gate before release)

✓ **Item**

- ☐ Build sanity passed (installs, opens, correct version, no launch crash) — on the **Play/Internal** build.
- ☐ All 4 **MUST-PASS** critical scenarios passed end-to-end.
- ☐ Regression (Section 3) done for all modules; new-feature deep-tested.
- ☐ **Zero open Sev-1 / Sev-2**. All logged in the bug tracker with severity.
- ☐ Tested on ≥2 devices, WiFi + mobile data.
- ☐ Post-release smoke + monitoring plan assigned (who watches, for how long).
- ☐ Rollback owner + kill-switches identified. Signed by QA lead + Tech lead.

## 5. AFTER RELEASE — production verification & monitoring AFTER

### 5.1 Smoke test on prod (first 30 min, 2–3 real devices)

|                          |                                                         |
|--------------------------|---------------------------------------------------------|
| ✓                        | Test on the LIVE app                                    |
| <input type="checkbox"/> | Fresh install from Play → open → login/OTP works.       |
| <input type="checkbox"/> | One real random call connects both ways.                |
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|                          |                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| ✓                        | Item                                                                                                         |
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| <input type="checkbox"/> | All 4 <b>MUST-PASS</b> critical scenarios passed end-to-end.                                                 |
| <input type="checkbox"/> | Regression (Section 3) done for all modules; new-feature deep-tested.                                        |
| <input type="checkbox"/> | <b>Zero open Sev-1 / Sev-2</b> . All logged in the bug tracker with severity.                                |
| <input type="checkbox"/> | Tested on ≥2 devices, WiFi + mobile data.                                                                    |
| <input type="checkbox"/> | Post-release smoke + monitoring plan assigned (who watches, for how long).                                   |
| <input type="checkbox"/> | Rollback owner + kill-switches identified. Signed by QA lead + Tech lead.                                    |

# Local Call-Audio Extraction Foundation

SOP revision 1.1 - 2026-07-17 - Applies to development APK before any transcription provider is connected.

## Release gate

ACX-001, 002, 003, 004, 005, 007 and 008 are MUST-PASS on real devices. A compile/unit-test pass alone does not authorize production rollout. Attach logs, profiler screenshots and defect/retest evidence.

| ACX-001 - Male local microphone extraction                   | MUST-PASS / P0                                                                                                                                                                                       |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preconditions / build / role                                 | Development APK; two real devices; male caller and female creator; call connects normally.                                                                                                           |
| Test data                                                    | Speak a fixed 65-second script on the male device; female remains quiet.                                                                                                                             |
| Steps                                                        | 1. Start an audio call. 2. Speak before and across 30 s and 60 s boundaries. 3. End normally. 4. Capture filtered AudioExtractor logs.                                                               |
| Expected result                                              | Call audio remains clear. Male device reports monotonic chunks with 30 s max and 1 s overlap after the first. Female local track does not contain male playback. No file/network/AI/warning appears. |
| Execution                                                    | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                    |
| ACX-002 - Female local microphone extraction and attribution | MUST-PASS / P0                                                                                                                                                                                       |
| Preconditions / build / role                                 | Same two-device build; connected audio call.                                                                                                                                                         |
| Test data                                                    | Female speaks fixed script while male stays quiet; then both alternate turns.                                                                                                                        |
| Steps                                                        | 1. Record female-only speech for 35 s. 2. Alternate speakers. 3. Compare each device's local metadata and audible call.                                                                              |
| Expected result                                              | Female device marks local speech. Male device marks only its own spoken turns, not remote playback. No speaker-attribution crossover or call-audio regression.                                       |
| Execution                                                    | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                    |
| ACX-003 - Manual mute privacy boundary                       | MUST-PASS / P0                                                                                                                                                                                       |
| Preconditions / build / role                                 | Connected call; metadata logs available.                                                                                                                                                             |
| Test data                                                    | Speak A while live, mute and speak PRIVATE, unmute and speak B.                                                                                                                                      |
| Steps                                                        | 1. Speak A. 2. Tap mute. 3. Speak for at least 10 s while muted. 4. Unmute and speak B. 5. End call.                                                                                                 |
| Expected result                                              | Peer hears A and B only. Pre-mute partial closes; no extracted duration advances during mute; post-unmute starts a clean non-overlapping boundary. PRIVATE is never present in any future payload.   |
| Execution                                                    | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                    |
| ACX-004 - Phone/audio-focus interruption                     | MUST-PASS / P0                                                                                                                                                                                       |
| Preconditions / build / role                                 | Connected call; second incoming cellular or VoIP/audio-focus event available.                                                                                                                        |
| Test data                                                    | Speak before, during and after the interruption.                                                                                                                                                     |
| Steps                                                        | 1. Trigger interruption. 2. Speak while HIMA is held. 3. End interruption. 4. Resume HIMA speech. 5. Repeat once.                                                                                    |
| Expected result                                              | Extraction pauses with outgoing mic, resumes with a clean boundary, sequence remains monotonic, call does not crash, duplicate workers are not created and held/private speech is excluded.          |
| Execution                                                    | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                    |



# Long-Call, VAD and Failure Coverage

Environment: development APK, two real devices, Wi-Fi and mobile data, wired/Bluetooth where available.

| ACX-005 - Boundary and hour-long stability           | MUST-PASS / P0                                                                                                                                                                                                              |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preconditions / build / role                         | Two charged test accounts; devices on power; Android Studio profiler or equivalent.                                                                                                                                         |
| Test data                                            | At least 65 min call; scripted words spanning each sampled 30 s boundary.                                                                                                                                                   |
| Steps                                                | 1. Run call for 65 min. 2. Sample logs at start, 30 min and 60 min. 3. Exercise speaker, wired and Bluetooth routes. 4. End normally.                                                                                       |
| Expected result                                      | No 30 s recording stop, missing tail, duplicate-only final chunk, ANR, OOM or increasing worker count. Windows stay <=960 KB; sequence/timestamps are monotonic; audible call quality remains normal.                       |
| Execution                                            | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                                           |
| ACX-006 - VAD shadow labels across languages         | P1                                                                                                                                                                                                                          |
| Preconditions / build / role                         | Connected call in a quiet room, then normal background noise.                                                                                                                                                               |
| Test data                                            | English plus all 11 HIMA languages and code-switching; silence; soft speech; loud speech; fan noise.                                                                                                                        |
| Steps                                                | 1. Speak each sample naturally. 2. Include quiet and noisy periods. 3. Compare hasSpeech/voiced ratio to the known speaking periods.                                                                                        |
| Expected result                                      | VAD is language-independent and labels ordinary speech without modifying/cutting PCM. Silence is normally false. Noise false positives are documented; any speech false negative blocks using VAD to skip future API calls. |
| Execution                                            | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                                           |
| ACX-007 - Teardown, reconnect and permission failure | MUST-PASS / P0                                                                                                                                                                                                              |
| Preconditions / build / role                         | Connected call; ability to toggle network and revoke RECORD_AUDIO.                                                                                                                                                          |
| Test data                                            | Wi-Fi/mobile switch, temporary loss, peer leave, local hangup, permission revoke.                                                                                                                                           |
| Steps                                                | 1. Reconnect network twice. 2. End via peer and local paths. 3. Repeat with permission revoke. 4. Start a fresh call after each case.                                                                                       |
| Expected result                                      | Observer is not duplicated on reconnect; dispose is safe from leaveChannel/onDestroy; no extractor thread survives teardown; next call captures normally; existing call failure handling remains authoritative.             |
| Execution                                            | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                                           |
| ACX-008 - Privacy, no-spend and regression gate      | MUST-PASS / P0                                                                                                                                                                                                              |
| Preconditions / build / role                         | Fresh development APK; proxy/file inspection tools; admin access for read-only verification.                                                                                                                                |
| Test data                                            | One 2-minute call containing harmless test speech.                                                                                                                                                                          |
| Steps                                                | 1. Inspect app private cache/database before/during/after. 2. Inspect network requests. 3. Check admin/warnings. 4. Verify call settlement/rating flow.                                                                     |
| Expected result                                      | Zero audio files, uploads, transcript rows, provider calls, AI cost, admin audio, moderation results or warnings. Existing connection, mute, timer, hangup, settlement and rating flows still complete.                     |
| Execution                                            | Status: NOT RUN   Evidence: _____   Defect: _____   Retest: _____                                                                                                                                                           |